

Precalculus Practice Problems: Topics 2.7–2.10

Name: _____

Date: _____

Directions: Show all work. Simplify completely. Give exact values unless stated otherwise.

Topic 2.7: Composition of Functions

2.7.1 Let $f(x) = 2x - 5$ and $g(x) = x^2 + 1$.

(a) Find $(f \circ g)(x)$ and simplify.

(b) Find $(g \circ f)(x)$ and simplify.

(c) Evaluate $(f \circ g)(-2)$.

2.7.2 Let $f(x) = \sqrt{x - 1}$ and $g(x) = 3x + 2$.

(a) Find $(f \circ g)(x)$.

(b) State the domain of $(f \circ g)(x)$.

2.7.3 Suppose $f(x) = \frac{1}{x-4}$ and $g(x) = x + 1$.

(a) Find $(f \circ g)(x)$ and state its domain.

(b) Find $(g \circ f)(x)$ and state its domain.

2.7.4 Let $f(x) = x^2$ and $h(x) = x - 3$.

(a) Write $(f \circ h)(x)$ in expanded form.

(b) Solve $(f \circ h)(x) = 16$.

FRQ 2.7: Composition

A factory records raw material input x (in tons). The cost of processing is modeled by $f(x) = 50x + 200$, and shipping cost is modeled by $g(x) = 0.1x$.

(a) Write a function that represents the total cost including shipping.

(b) Evaluate the total cost when $x = 30$ tons.

- (c) Explain the meaning of $(g \circ f)(x)$ in this context and state whether it is meaningful.

Topic 2.8: Inverse Functions

2.8.1 Find the inverse of $f(x) = 5x - 7$.

2.8.2 Determine whether $f(x) = x^2 - 4$ has an inverse on $\mathbb{R}$. If not, restrict the domain and find the inverse.

2.8.3 Given $f(x) = \frac{x-2}{3}$:

(a) Find $f^{-1}(x)$.

(b) Verify that $f(f^{-1}(x)) = x$.

2.8.4 A function contains the points $(1, 4)$, $(2, 7)$, and $(5, 13)$.

(a) Write three points on f^{-1} .

(b) Describe how the graphs of f and f^{-1} are related.

FRQ 2.8: Inverses

A ride-sharing company charges a base fee of \$6 plus \$1.50 per mile traveled.

(a) Write a function that models the total cost in terms of miles driven.

(b) Find the inverse function.

(c) Explain what the inverse represents in this situation.

Topic 2.9: Logarithmic Expressions

2.9.1 Rewrite the expression using properties of logarithms:

$$\log_4(64x^2)$$

2.9.2 Expand completely:

$$\ln \left(\frac{x^3 \sqrt{y}}{5} \right)$$

2.9.3 Condense into a single logarithm:

$$\log_2(x) + \log_2(y) - \log_2(4)$$

2.9.4 Solve for x :

$$\log_3(x) + \log_3(x - 2) = 2$$

FRQ 2.9: Logarithms

The pH of a solution is given by

$$\text{pH} = -\log_{10}[H^+]$$

- (a) Find the pH of a solution with $[H^+] = 10^{-6}$.

(b) A solution has a pH of 4. Find its hydrogen ion concentration.

(c) Explain why the pH scale is logarithmic rather than linear.

Topic 2.10: Inverses of Exponential Functions

2.10.1 Find the inverse of $f(x) = 2^x$.

2.10.2 Find the inverse of $g(x) = 3 \cdot 5^x$ and evaluate $g^{-1}(75)$.

2.10.3 Solve exactly:

$$4^{x-1} = 32$$

2.10.4 Let $h(x) = e^{x+2}$.

(a) Find $h^{-1}(x)$.

- (b) State the domain and range of h and h^{-1} .

FRQ 2.10: Exponential Models

A bacteria culture grows according to

$$P(t) = 500(2)^t$$

where t is measured in hours.

- (a) Find an expression for t in terms of P .
- (b) How long does it take for the population to reach 4000 bacteria?
- (c) Explain why logarithms are necessary to solve this problem.